1

DA TUITION

BEYOND ACADEMIC EXCELLENCE

2 Unit BOS

Calculus II

Curve Sketching

Name: **Solutions**

Question 1

1976

Find the greatest and least values of y where $y = x^3 - 12x + 20$ and x is restricted to the interval $-3 < x < 5$.

(ii) Let $y = f(x) = x^3 - 12x + 20$

$$f'(x) = 3x^2 - 12$$
$$= 3(x^2 - 4)$$
$$= 3(x+2)(x-2)$$
$$f''(x) = 6x$$

Stationary points occur when $f'(x) = 0$

i.e. $f'(x) = 3(x+2)(x-2) = 0$

$\therefore x = \pm 2$

at $x = 2$ $y = 4$, at $x = -2$ $y = 36$

$f''(2) = (+)(+) > 0$

$\therefore$ A minimum turning point at $(2, 4)$

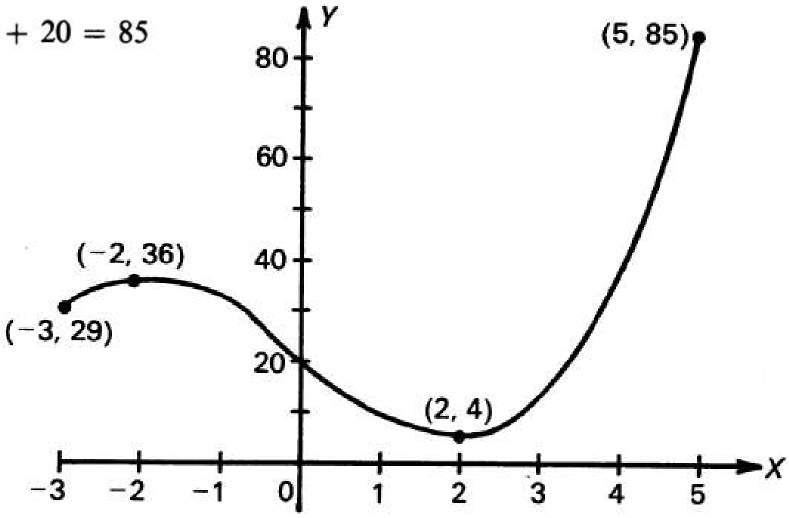
$f''(-2) = (+)(-) < 0$

$\therefore$ A maximum turning point at $(-2, 36)$

Values at end points:

at $x = -3$ $y = (-3)^3 - 12(-3) + 20$
$= 29$

at $x = 5$ $y = (5)^3 - 12(5) + 20 = 85$



$\therefore$ For $-3 \leqslant x \leqslant 5$

Greatest value $= 85$

Least value $= 4$.

Out of

Question 2 1978

For the curve

$$y = x^3 - 3x^2 - 12$$

find the stationary points and determine their nature. Also find any points of inflexion.

$$\text{Let } y = f(x) = x^3 - 3x^2 - 12$$
$$f(x) = 3x^2 - 6x$$
$$= 3x(x - 2)$$
$$f''(x) = 6x - 6$$
$$= 6(x - 1)$$

Stationary points occur when $f'(x) = 0$

$$f'(x) = 3x(x - 2) = 0$$
$$\therefore x = 0, 2.$$

at $x = 0$ $y = -12$, at $x = 2$ $y = -16$.

$$f''(0) = (+)(-) < 0$$

$\therefore$ A maximum turning point at $(0, -12)$

$$f''(2) = (+)(+) > 0$$

$\therefore$ A minimum turning point at $(2, -16)$

Points of inflexion occur when $f''(x) = 0$

$$f''(x) = 6(x - 1) = 0$$
$$\therefore x = 1.$$
$$f''(1 - \varepsilon) = (+)(-) < 0$$
$$f''(1 + \varepsilon) = (+)(+) > 0$$

$\therefore$ A point of inflexion at $(1, -14)$.

Out of

Question 3

1983

The function $f(x)$ is defined by the rule

$f(x) = 5x\sqrt{(x+1)}$,

in the domain $-1 < x < 1$.

(a) Draw up a table of values of $f(x)$, correct to one decimal place, for each of the values $x = -1, -\frac{1}{2}, 0, \frac{1}{2}, 1$.

(b) Use the derivative of $f(x)$ to find the turning point of $f(x)$.

(c) Hence draw a sketch of $f(x)$ showing clearly the turning point, and the values at the end points of the domain.

(a)

x	-1	$-^1/_2$	0	$^1/_2$	1
y	0	-1.8	0	3.1	7.1

(b)
$$\begin{aligned} f(x) &= 5x\sqrt{x+1} \\ &= 5x(x+1)^{1/2} \\ f'(x) &= 5x.\frac{d}{dx}(x+1)^{1/2} + (x+1)^{1/2}.\frac{d}{dx}(5x) \\ &= 5x.\tfrac{1}{2}(x+1)^{-1/2}.1 + (x+1)^{1/2}.5 \\ &= \frac{5x}{2\sqrt{x+1}} + 5\sqrt{x+1} \\ &= \frac{5x + (5\sqrt{x+1})(2\sqrt{x+1})}{2\sqrt{x+1}} \\ &= \frac{5x + 10(x+1)}{2\sqrt{x+1}} \\ &= \frac{15x+10}{2\sqrt{x+1}} \\ &= \frac{15(x+{}^2/_3)}{2\sqrt{x+1}} \end{aligned}$$

Stationary points occur when $f'(x) = 0$

$$\frac{15(x+{}^2/_3)}{2\sqrt{x+1}} = 0$$

$$15(x+{}^2/_3) = 0$$

$$x = \frac{-2}{3}$$

$$f\left(\frac{-2}{3}\right) = \left(\frac{-10}{3}\right)\sqrt{\frac{1}{3}}$$

$= -1.9$ to 1 decimal place (by calc.)

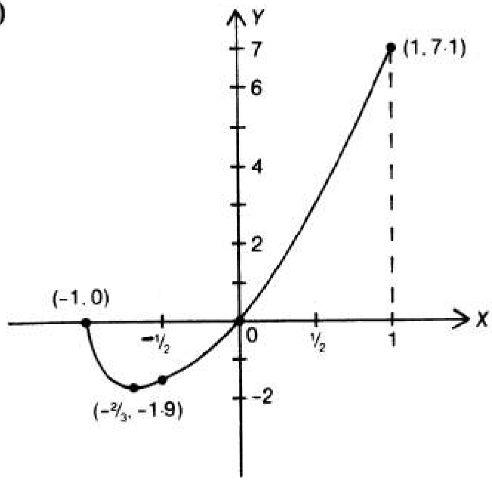
To determine the nature of the stationary point:

$$f'\left(\tfrac{-2}{3} - \epsilon\right) = \frac{(+)(-)}{(+)} < 0$$

$$f'\left(\tfrac{-2}{3} + \epsilon\right) = \frac{(+)(+)}{(+)} > 0$$

$\therefore$ A minimum turning point at $({}^{-2}/_3, -1.9)$.
(Alternatively, $f'({}^{-3}/_4)$ and $f'({}^{-1}/_2)$ could be used.)

(c)



Out of

Question 4

For the domain $-1 \leqslant x \leqslant 3$, determine the maximum value and the minimum value of each of the functions defined by

$$f(x) = 3x^2, \qquad g(x) = 4x - x^2.$$

Find all values of x in the interval $-1 \leqslant x \leqslant 3$ for which $f(x) = g(x)$.

Sketch the curves $y = f(x)$ and $y = g(x)$, for $-1 \leqslant x \leqslant 3$, on the same diagram. Indicate clearly the points where maximum and minimum values are attained and the points of intersection of the two curves.

Stationary points occur when $f'(x) = 0$ and $g'(x) = 0$

$f(x) = 3x^2$ $\quad$ $g(x) = 4x - x^2$

$f'(x) = 6x$ $\quad$ $g'(x) = 4 - 2x$

$f''(x) = 6$ $\quad$ $g''(x) = -2$

$f'(x) = 0$

when $6x = 0$

$\therefore x = 0$

$f(0) = 3(0)^2 = 0$

$f''(0) = 6 > 0$

$\therefore$ A minimum turning point at (0, 0)

$g'(x) = 0$

when $4 - 2x = 0$

$\therefore x = 2$

$g''(2) = -2 < 0$

$\therefore$ A maximum turning point at (2, 4)

Test end points of the domain:

x	-1	3
$f(x)$	3	27
$g(x)$	-5	3

For $f(x)$:
Maximum value is 27 when $x = 3$
Minimum value is 0 when $x = 0$
For $g(x)$:
Maximum value is 4 when $x = 2$
Minimum value is -5 when $x = -1$

If $f(x) = g(x)$

$3x^2 = 4x - x^2$

$4x^2 - 4x = 0$

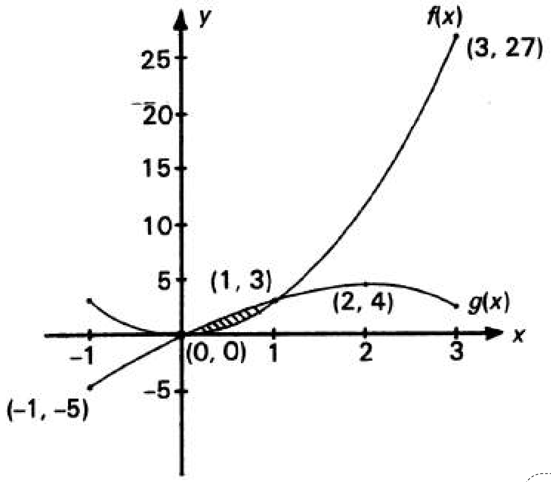
$4x(x - 1) = 0$

$\therefore x = 0$ or $x = 1$

$f(0) = 3(0)^2 = 0$

$f(1) = 3(1)^2 = 3$

Points of intersection are (0, 0) and (1, 3)



Out of

Question 5 1985

The curve

$$y = 3x^2 + \frac{a}{x^2}$$

has a turning point at $x = 3$. Find the constant a.

(iv) $y = 3x^2 + \dfrac{a}{x^2}$

$= 3x^2 + ax^{-2}$

$\dfrac{dy}{dx} = 6x - 2ax^{-3}$

$= 6x - \dfrac{2a}{x^3}$

At the turning point when $x = 3$,

$$\frac{dy}{dx} = 0$$

$$\therefore\ 6(3) - \frac{2a}{3^3} = 0$$

$$18 = \frac{2a}{27}$$

$$2a = 18 \times 27$$

$$a = \frac{18 \times 27}{2}$$

$$= 243.$$

Out of

Question 6

1991

Consider the curve given by $y = 3x^2 - x^3$.

(i) Find the stationary points and determine their nature.

(ii) Sketch the curve, indicating where it crosses the x axis.

(i) $y = 3x^2 - x^3$

$$\frac{dy}{dx} = 6x - 3x^2$$
$$= 3x(2 - x)$$
$$\frac{d^2y}{dx^2} = 6 - 6x$$
$$= 6(1 - x)$$

Stationary points occur when

$$\frac{dy}{dx} = 0$$
$$3x(2 - x) = 0$$
$$\therefore x = 0 \text{ or } 2$$

When $x = 0$, $y = 3(0)^2 - (0)^3$
$= 0$

and $\dfrac{d^2y}{dx^2} = 6(1 - 0) > 0$

$\therefore$ A minimum turning point at (0, 0).

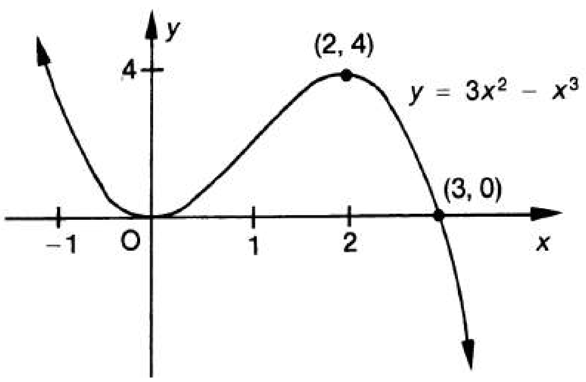
When $x = 2$, $y = 3(2)^2 - (2)^3$
$= 12 - 8$
$= 4$

and $\dfrac{d^2y}{dx^2} = 6(1 - 2) < 0$

$\therefore$ A maximum turning point at (2, 4)

(ii) Curve crosses x axis when $y = 0$,

$$\therefore 3x^2 - x^3 = 0$$
$$x^2(3 - x) = 0$$
$$\therefore x = 0 \text{ or } 3$$



Out of

Question 7 | 1987

The function $f(x)$ is given by

$$f(x) = x(x-3)^2.$$

(a) Find the coordinates of the points at which the graph of $y = f(x)$ meets the x-axis.

(b) Find the coordinates of the turning points of $f(x)$ and state whether they are maxima or minima.

(c) Draw a sketch of $y = f(x)$ in the domain $-1 \leqslant x \leqslant 4$.

(a) The graph meets the x-axis where $y = 0$

i.e. $f(x) = 0$

$\therefore\ x(x-3)^2 = 0$

$\therefore\ x = 0$

or $(x-3)^2 = 0$

$\therefore\ x = 3$

$\therefore$ points are (0, 0) and (3, 0).

(b)

$$\begin{aligned} f(x) &= x(x-3)^2 \\ &= x(x^2 - 6x + 9) \\ &= x^3 - 6x^2 + 9x \\ f'(x) &= 3x^2 - 12x + 9 \\ &= 3(x^2 - 4x + 3) \\ &= 3(x-3)(x-1) \\ f''(x) &= 6x - 12 \\ &= 6(x-2) \end{aligned}$$

Stationary points occur when

$f'(x) = 0$

$3(x-3)(x-1) = 0$

$\therefore x = 3 \text{ or } 1$

$f(3) = 3(3-3)^2 = 0$

$f(1) = 1(1-3)^2 = 4$

$f''(3) = 6(3-2) \quad >0$

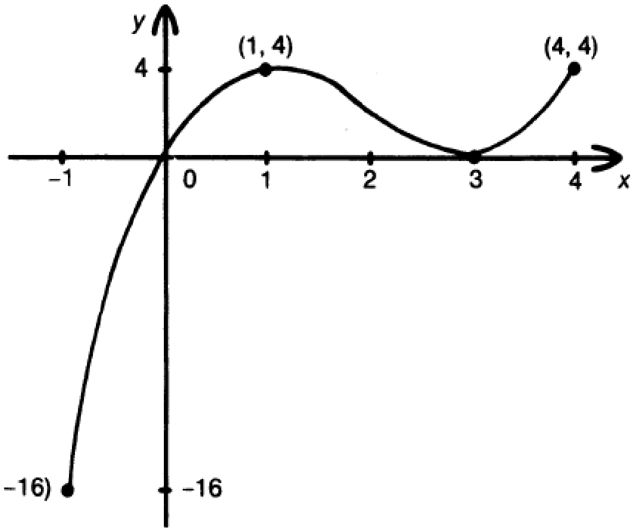
$\therefore$ A minimum turning point at (3, 0)

$f''(1) = 6(1-2) \quad <0$

$\therefore$ A maximum turning point at (1, 4).

(c) Values at end points of domain:

$$\begin{aligned} f(-1) &= -1(-1-3)^2 \\ &= -1(16) \\ &= -16 \\ f(4) &= 4(4-3)^2 \\ &= 4(1) \\ &= 4 \end{aligned}$$



Out of

Question 8 **1993**

Consider the curve given by $y = \frac{1}{4}x^4 - x^3$.

(i) Find any turning points and determine their nature.

(ii) Find any points of inflexion.

(iii) Sketch the curve for $-1{\cdot}5 \le x \le 4{\cdot}5$, indicating where the curve crosses the x axis.

(iv) For what values of x is the curve concave down?

(a) (i) $y = \frac{1}{4}x^4 - x^3$
$= x^3(\frac{1}{4}x - 1)$

$\frac{dy}{dx} = x^3 - 3x^2$
$= x^2(x - 3)$

$\frac{d^2y}{d^2x} = 3x^2 - 6x$
$= 3x(x - 2)$

Stationary points occur when
$\frac{dy}{dx} = 0$
$x^2(x - 3) = 0$
$\therefore x = 0$ or 3

When $x = 0, y = 0$
and $\frac{d^2y}{dx^2} = 0$

$\therefore$ Possible point of inflexion or turning point at $(0, 0)$.

When $x = 0 - \epsilon$, $\frac{dy}{dx} = (+)(-) < 0$

When $x = 0 + \epsilon$, $\frac{dy}{dx} = (+)(-) < 0$

$\therefore$ A horizontal point of inflexion at $(0, 0)$.

When $x = 3, y = \frac{1}{4}(3)^4 - (3)^3$
$= \frac{81}{4} - 27$
$= \frac{81}{4} - \frac{108}{4}$
$= -\frac{27}{4}$
$= -6\frac{3}{4}$

and $\frac{d^2y}{dx^2} = 3(3)(1) > 0$

$\therefore$ A minimum turning point at $(3, -6\frac{3}{4})$.

(ii) Points of inflexion may occur when
$\frac{d^2y}{dx^2} = 0$
i.e. $3x(x - 2) = 0$
$\therefore x = 0$ or 2

From (i), there is a horizontal point of inflexion at $(0, 0)$.

When $x = 2, y = \frac{1}{4}(2)^4 - (2)^3$
$= \frac{16}{4} - 8$
$= 4 - 8$
$= -4$

When $x = 2 - \epsilon$, $\frac{d^2y}{dx^2} = (+)(+)(-) < 0$

When $x = 2 + \epsilon$, $\frac{d^2y}{dx^2} = (+)(+)(+) > 0$

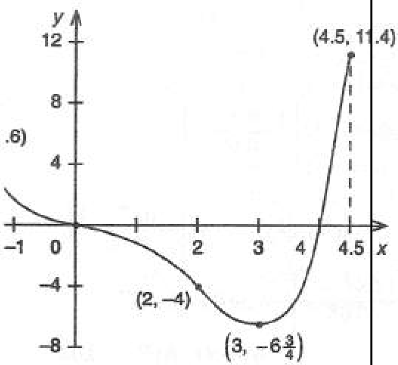
$\therefore$ A point of inflexion at $(2, -4)$.

(iii) Values at end points:

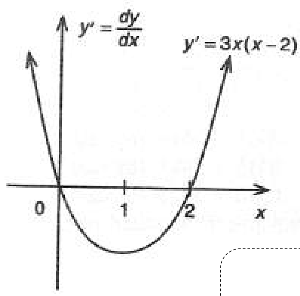
When $x = -1.5$,
$y = \frac{1}{4}(-1.5)^4 - (-1.5)^3$
$= \frac{1}{4}(5.0625) - (-3.375)$
$= 4.6406\ldots$ (by calc.)
$= 4.6$ to 1 decimal place

When $x = 4.5$,
$y = \frac{1}{4}(4.5)^4 - (4.5)^3$
$= \frac{1}{4}(410.0625) - (91.125)$
$= 11.3906\ldots$ (by calc.)
$= 11.4$ to 1 decimal place

The curve crosses the x-axis when
$y = 0$
i.e. $x^3(\frac{1}{4}x - 1) = 0$
$\therefore x = 0$ or 4



The curve is concave down when
$\frac{d^2y}{dx^2} < 0$
i.e. $3x(x - 2) < 0$



$\therefore 0 < x < 2$

Out of

Question 9

1994

Consider the curve given by $y = x^3 - 6x + 4$.

(i) Find the coordinates of the stationary points and determine their nature.

(ii) Find the coordinates of any point of inflexion.

(iii) Sketch the curve for the domain $-3 \le x \le 3$.

(iv) What is the maximum value of $x^3 - 6x + 4$ in the domain $-3 \le x \le 3$?

(c) (i) $y = x^3 - 6x + 4$

$$\frac{dy}{dx} = 3x^2 - 6$$
$$= 3(x^2 - 2)$$
$$= 3(x + \sqrt{2})(x - \sqrt{2})$$
$$\frac{d^2y}{dx^2} = 6x$$

Stationary points occur when

$$\frac{dy}{dx} = 0$$
$$3(x + \sqrt{2})(x - \sqrt{2}) = 0$$
$$\therefore x = \pm\sqrt{2}$$

When $x = \sqrt{2}$, $y = (\sqrt{2})^3 - 6(\sqrt{2}) + 4$
$$= 2\sqrt{2} - 6\sqrt{2} + 4$$
$$= 4 - 4\sqrt{2}$$

and $\frac{d^2y}{dx^2} = 6(\sqrt{2}) > 0$

$\therefore$ A minimum turning point at $(\sqrt{2}, 4 - 4\sqrt{2})$.

When $x = -\sqrt{2}$,
$$y = (-\sqrt{2})^3 - 6(-\sqrt{2}) + 4$$
$$= -2\sqrt{2} + 6\sqrt{2} + 4$$
$$= 4 + 4\sqrt{2}$$

and $\frac{d^2y}{dx^2} = 6(-\sqrt{2}) < 0$

$\therefore$ A maximum turning point at $(-\sqrt{2}, 4 + 4\sqrt{2})$.

(ii) Points of inflexion may occur when

$$\frac{d^2y}{dx^2} = 0$$

i.e. $6x = 0$

$\therefore x = 0$

When $x < 0$, say $x = -\frac{1}{2}$,
$$\frac{d^2y}{dx^2} = 6(-\tfrac{1}{2}) < 0$$

When $x > 0$, say $x = \frac{1}{2}$
$$\frac{d^2y}{dx^2} = 6(\tfrac{1}{2}) > 0$$

$\therefore$ A point of inflexion at (0, 4)

(iii) Values at end points of domain:

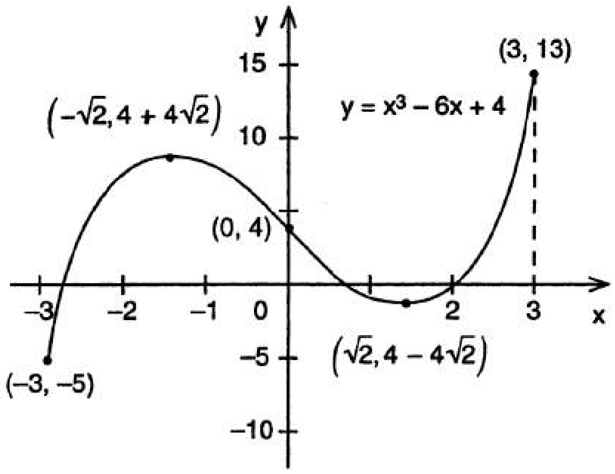
At $x = -3$, $y = (-3)^3 - 6(-3) + 4$
$$= -27 + 18 + 4$$
$$= -5$$

At $x = 3$, $y = (3)^3 - 6(3) + 4$
$$= 27 - 18 + 4$$
$$= 13$$

Y intercept:

At $x = 0$, $y = 0 - 0 + 4$
$$= 4$$

i.e. (0, 4)



(iv) Maximum value is 13.

Out of

Question 10 1995

Consider the curve given by $y = 7 + 4x^3 - 3x^4$.

(i) Find the coordinates of the two stationary points.

(ii) Find all values of x for which $\frac{d^2y}{dx^2} = 0$.

(iii) Determine the nature of the stationary points.

(iv) Sketch the curve for the domain $-1 \le x \le 2$.

(i) $y = 7 + 4x^3 - 3x^4$

$$\frac{dy}{dx} = 12x^2 - 12x^3 = 12x^2(1 - x)$$

Stationary points occur when

$$\frac{dy}{dx} = 0$$

$12x^2(1 - x) = 0$

$\therefore x = 0$ or 1

When $x = 0$, $y = 7 + 0 - 0 = 7$

When $x = 1$, $y = 7 + 4 - 3 = 8$

$\therefore$ The coordinates of the stationary points are $(0, 7)$ and $(1, 8)$.

(ii) $$\frac{d^2y}{dx^2} = 24x - 36x^2 = 12x(2 - 3x)$$

Now $\frac{d^2y}{dx^2} = 0$

when $12x(2 - 3x) = 0$

$\therefore x = 0$ or $\frac{2}{3}$.

(iii) When $x = 0$, $\frac{d^2y}{dx^2} = 0$

When $x < 0$, say $x = -0.1$

$$\frac{d^2y}{dx^2} = 24(-0.1) - 36(-0.1)^2 = -2.43$$

When $x > 0$, say $x = +0.1$

$$\frac{d^2y}{dx^2} = 24(0.1) - 36(0.1)^2 = 2.04$$

$\therefore$ A horizontal point of inflexion at $(0, 7)$.

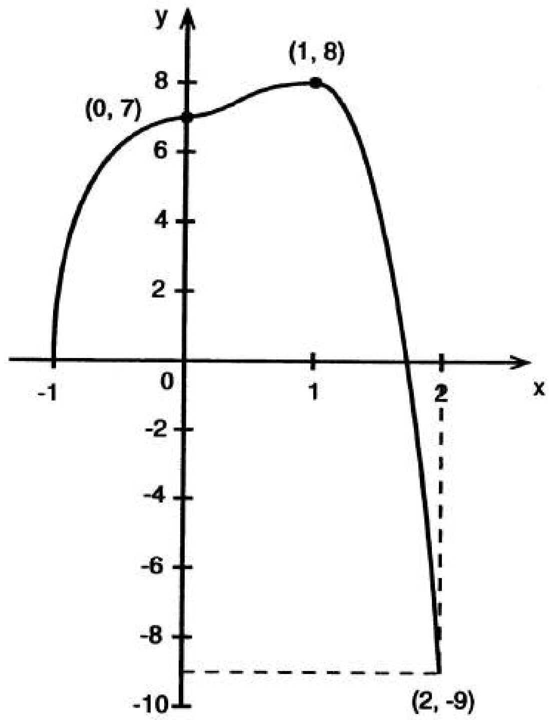
When $x = 1$, $\frac{d^2y}{dx^2} = 12(1)(-1) < 0$

$\therefore$ A maximum turning point at $(1, 8)$.

(iv) Values at end points of domain:

At $x = -1$, $y = 7 + 4(-1)^3 - 3(-1)^4 = 7 - 4 - 3 = 0$

At $x = 2$, $y = 7 + 4(2)^3 - 3(2)^4 = 7 + 32 - 48 = -9$



Out of

Question 11 2003

Consider the function $f(x)=x^4-4x^3$.

(i) Show that $f'(x)=4x^2(x-3)$.

(ii) Find the coordinates of the stationary points of the curve $y=f(x)$, and determine their nature.

(iii) Sketch the graph of the curve $y=f(x)$, showing the stationary points.

(iv) Find the values of x for which the graph of $y=f(x)$ is concave down.

(a)(i) $f(x) = x^4 - 4x^3$
$f'(x) = 4x^3 - 12x^2$
$= 4x^2(x-3)$ ✓

(ii) $f'(x) = 0$
$4x^2(x-3) = 0$
$x = 0 \quad x = 3$ ✓
$y = 0 \quad y = -27$

test $f''(x)$
$f''(x) = 12x^2 - 24x$
sub in $x = 0$
$0,\ 0 \ \therefore$ ✓
$\therefore$ horizontal p.of.i.

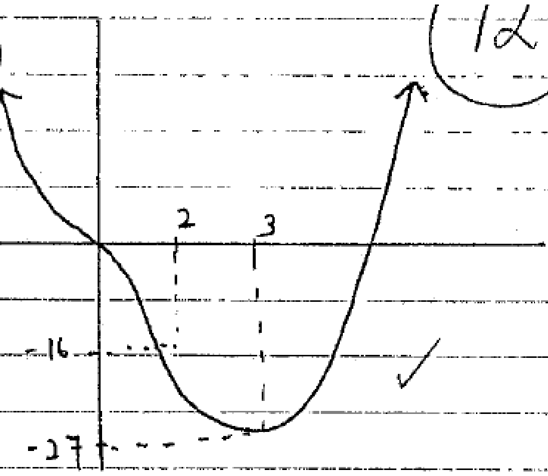
sub in $x = 3$
$12(3)^2 - 72$ ✓
$= 36 > 0 \ \therefore$ min.

$f''(x) < 0$
$12x^2 - 24x < 0$
$12x(x-2) < 0$

$x = 2$
$x = 0$ ✓

sub in 2.
$(2)^4 - 4(2)^3$
$= -16$

(iii) 12



✓

(iv) $0 < x < 2$ ✓

Out of

Question 12

2005

A function $f(x)$ is defined by $f(x) = (x+3)(x^2-9)$.

(i) Find all solutions of $f(x) = 0$.

(ii) Find the coordinates of the turning points of the graph of $y = f(x)$, and determine their nature.

(iii) Hence sketch the graph of $y = f(x)$, showing the turning points and the points where the curve meets the x-axis.

(iv) For what values of x is the graph of $y = f(x)$ concave down?

(i) $f(x) = (x+3)(x^2-9) = 0$

$$(x+3)(x-3)(x+3) = 0$$

$$\therefore \quad x = -3,\ 3.$$

(ii) $f(x) = x(x^2-9) + 3(x^2-9)$

$$= x^3 - 9x + 3x^2 - 27.$$

$$f'(x) = 3x^2 - 9 + 6x.$$

Stationary points when $f'(x) = 0$:

$$3x^2 + 6x - 9 = 0$$

$$3(x^2 + 2x - 3) = 0$$

$$3(x+3)(x-1) = 0$$

$$\therefore \quad x = -3 \text{ or } 1.$$

When $x = -3$, $y = (-3+3)\left[(-3)^2 - 9\right]$

$$= 0.$$

$\therefore$ Stationary point at $(-3, 0)$.

When $x = 1$, $y = (1+3)(1^2 - 9)$

$$= -32.$$

$\therefore$ Stationary point at $(1, -32)$.

To determine the nature of the turning points, the second derivative can be used.

$$f''(x) = 6x + 6.$$

Test $x = -3$: $f''(-3) = -18 + 6$

$$= -12.$$

ie. $f''(-3) < 0 \Rightarrow$ concave down

$\therefore$ Maximum turning point at $(-3, 0)$.

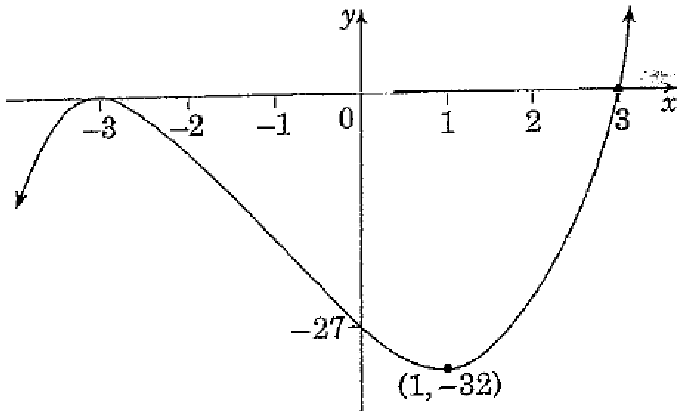
Test $x = 1$: $f''(1) = 6 + 6$

$$= 12.$$

ie. $f''(1) > 0 \Rightarrow$ concave up

$\therefore$ Minimum turning point at $(1, 32)$.

(iii) x-intercepts found by solving $f(x) = 0$ from (i), x-intercepts at -3 and 3.



(iv) $f(x)$ concave down when $f''(x) < 0$,

ie. $6x + 6 < 0$

$$6x < -6.$$

$\therefore$ Concave down when $x < -1$.

Out of

Question 13

2006

A function $f(x)$ is defined by $f(x) = 2x^2(3-x)$.

(i) Find the coordinates of the turning points of $y = f(x)$ and determine their nature.

(ii) Find the coordinates of the point of inflexion.

(iii) Hence sketch the graph of $y = f(x)$, showing the turning points, the point of inflexion and the points where the curve meets the x-axis.

(iv) What is the minimum value of $f(x)$ for $-1 \le x \le 4$?

(i) $f(x) = 2x^2(3-x)$
$= 6x^2 - 2x^3$
$f'(x) = 12x - 6x^2$
$f''(x) = 12 - 12x$

Stationary points occur when $f'(x) = 0$

$\therefore \quad 12x - 6x^2 = 0$
$6x(2-x) = 0$
$\therefore \; x = 0,\ x = 2.$

To determine the nature of turning points, the second derivative may be used.

When $x = 0$, $f(0) = 6(0)^2 - 2(0)^3$
$= 0$
and $f''(x) = 12 - 12(0)$
$= 12 > 0 \Rightarrow$ concave up

$\therefore$ Minimum turning point at $(0, 0)$.

When $x = 2$, $f(2) = 6(2)^2 - 2(2)^3$
$= 8$
and $f''(2) = 12 - 12(2)$
$= -12 < 0 \Rightarrow$ concave down

$\therefore$ Maximum turning point at $(2, 8)$.

(ii) Point of inflexion occurs when $f''(x) = 0$ and concavity changes.

$12 - 12x = 0$
$12 = 12x$
$x = 1,$
$f(1) = 6(1)^2 - 2(1)^3$
$= 4.$

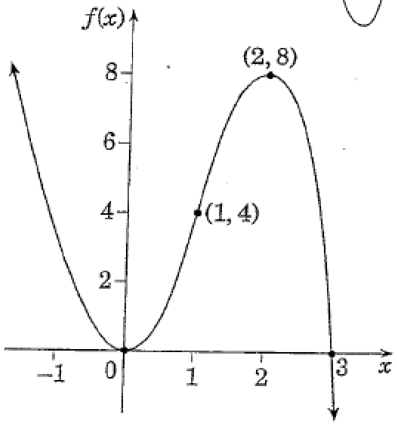
Considering concavity:

x	$\frac{1}{2}$	1	$1\frac{1}{2}$
$f''(x)$	$+6$	0	-6

$\therefore$ Concavity changes from positive to negative at $x = 1$.

$\therefore$ Point of inflexion at $(1, 4)$.

(iii) Note that the function has the shape of a negative cubic.



x-intercepts occur when $f(x) = 0$

$\therefore \quad 2x^2(3-x) = 0.$

$\therefore \; x = 0,\ x = 3$ are x-intercepts.

(iv) When restricting the domain to $-1 \le x \le 4$, the graph will have a minimum value at $x = 4$.

$\therefore$ Minimum value $= f(4)$
$= 6(4)^2 - 2(4)^3$
$= -32.$

Out of

Question 14

2007

Let $f(x) = x^4 - 4x^3$.

(i) Find the coordinates of the points where the curve crosses the axes. **2**

(ii) Find the coordinates of the stationary points and determine their nature. **4**

(iii) Find the coordinates of the points of inflexion. **1**

(iv) Sketch the graph of $y = f(x)$, indicating clearly the intercepts, stationary points and points of inflexion. **3**

(b)(i) $f(x)=x^4-4x^3$

$0=x^3(x-4)$

$\therefore$ cuts x axis at $(0,0)$ and $(4,0)$

(ii) $f'(x)=4x^3-12x^2$

$0=4x^2(x-3)$

st. points are $(0,0)$ and $(3,0)$

x	-1	0	1
$f'(x)$	$-$	0	$-$

$\therefore$ hort.p;oint of inflection at $(0,0)$

x	2	3	4
$f'(x)$	-	0	+

$\therefore$ rel. min at $(3,-27)$

(iii)

$f''(x)=12x^2-24x$

$0=12x(x-2)$

x=0 is a hort. point of inflection

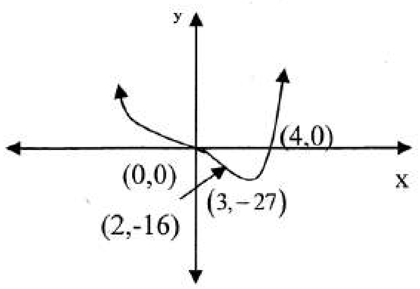
x=2

x	1	2	3
$f''(x)$	-	0	+

$\therefore$ change of concavity

$\therefore$ point of inflection at $(2,-16)$

(iv)



Out of

Question 15

2008

Let $f(x) = x^4 - 8x^2$.

(i) Find the coordinates of the points where the graph of $y = f(x)$ crosses the axes.

(ii) Show that $f(x)$ is an even function.

(iii) Find the coordinates of the stationary points of $f(x)$ and determine their nature.

(iv) Sketch the graph of $y = f(x)$.

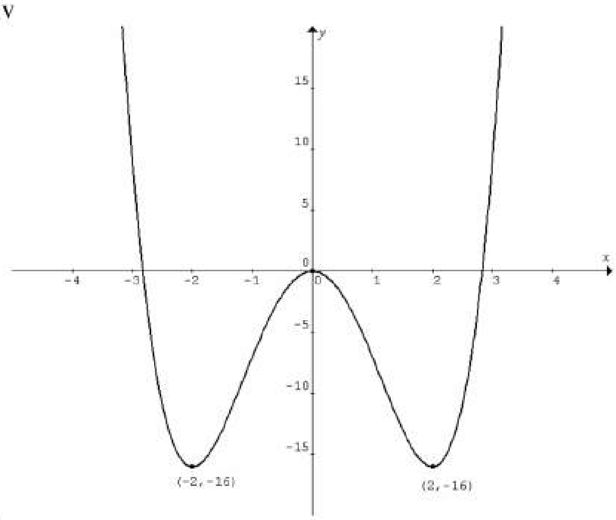
Q8ai $f(x)=0$, $x^4-8x^2=0$, $x^2(x^2-8)=0$. $\therefore x=0$ or $\pm 2\sqrt{2}$. The graph crosses the y-axis at $(0,0)$, or crosses the x-axis at $(-2\sqrt{2},0)$ or $(2\sqrt{2},0)$.

Q8aii $f(x)=x^4-8x^2$,
$f(-x)=(-x)^4-8(-x)^2=x^4-8x^2=f(x)$,
$\therefore f(x)$ is an even function.

Q8aiii $f'(x)=4x^3-16x=0$, $4x(x^2-4)=0$, $\therefore x=0$ or ± 2.

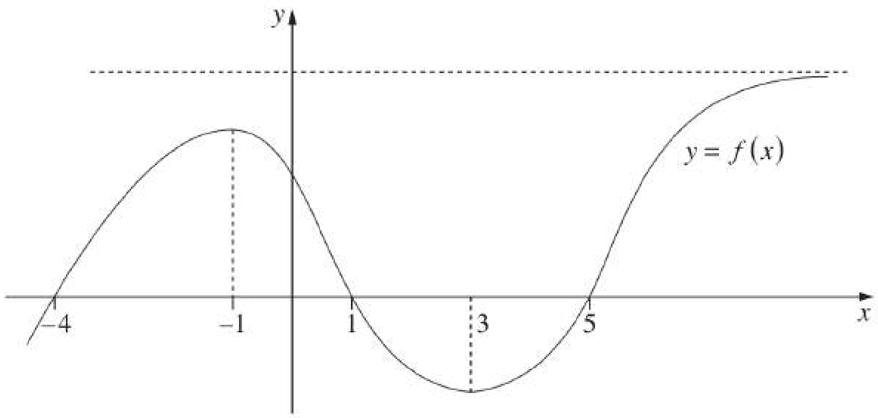
x		-2		0		2	
y		-16		0		-16	
(x,y)		$(-2,-16)$		$(0,0)$		$(2,-16)$	
dy/dx	$-$	0	$+$	0	$+$	0	$-$
Nature		Local min		Local max		Local min	

Q8aiv



Out of

Question 16 2009



The diagram shows the graph of a function $y = f(x)$.

(i) For which values of x is the derivative, $f'(x)$, negative? **1**

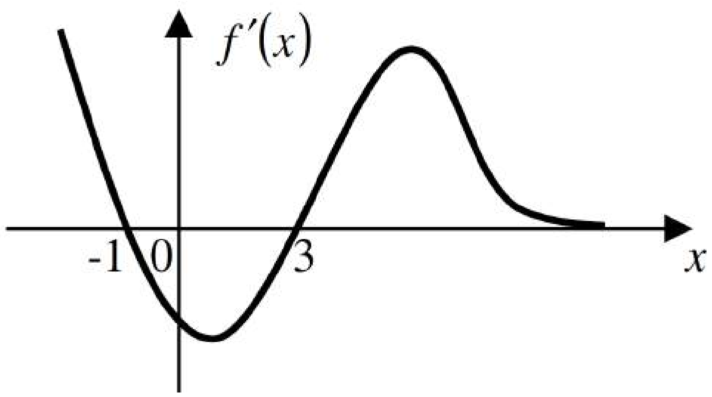
(ii) What happens to $f'(x)$ for large values of x? **1**

(iii) Sketch the graph $y = f'(x)$. **2**

Q8ai $-1 < x < 3$

Q8aii As $x \to \infty$, $f'(x) \to 0^+$.

Q8aiii $f'(x) = 0$ when $x = -1, 3$ or $x \to \infty$.



Out of

Question 17

2010

Let $f(x) = (x+2)(x^2+4)$.

(i) Show that the graph $y = f(x)$ has no stationary points.

(ii) Find the values of x for which the graph $y = f(x)$ is concave down, and the values for which it is concave up.

(iii) Sketch the graph $y = f(x)$, indicating the values of the x and y intercepts.

Q6ai $f(x) = (x+2)(x^2+4)$,

$f'(x) = (x+2)(2x) + (x^2+4) = 3x^2 + 4x + 4 \neq 0$ because

$\Delta = 4^2 - 4(3)(4) < 0$

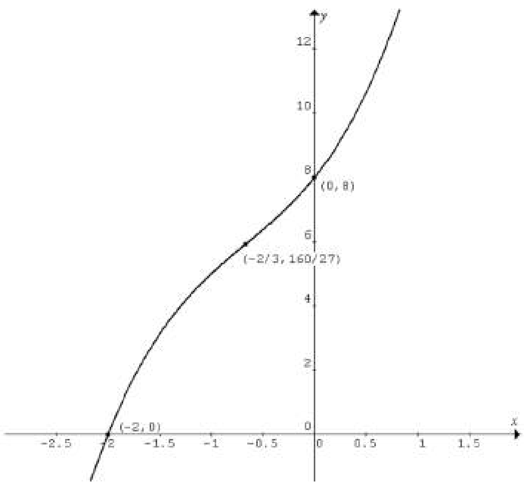
.: $f(x)$ has no stationary points.

Q6aii $f''(x) = 6x + 4 = 0$, $x = -\dfrac{2}{3}$.

The graph of $y = f(x)$ is concave down for $x < -\dfrac{2}{3}$, and it is concave up for $x > -\dfrac{2}{3}$.

Q6aiii x-intercept: Let $y = 0$, $(x+2)(x^2+4) = 0$, .: $x = -2$

y-intercept: Let $x = 0$, $y = (0+2)(0^2+4) = 8$



Out of

Question 18

2011

Let $f(x) = x^3 - 3x + 2$.

(i) Find the coordinates of the stationary points of $y = f(x)$, and determine their nature.

(ii) Hence, sketch the graph $y = f(x)$ showing all stationary points and the y-intercept.

Q7ai $f(x) = x^3 - 3x + 2$, $f'(x) = 3x^2 - 3 = 3(x+1)(x-1)$

$f'(x) = 0$ at $x = -1$, $y = 4$ or $x = 1$, $y = 0$

Stationary points are $(-1,4)$ and $(1,0)$

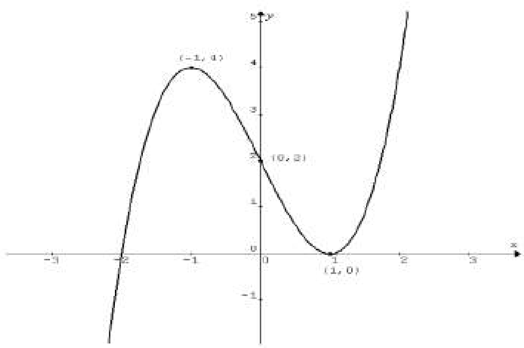
Since the coefficient of the x^3 term is a positive value, the shape of $f(x)$ is .

OR

x	-2	-1	0	1	2
$f'(x)$	$+$	0	$-$	0	$+$

.: $(-1,4)$ is a local maximum and $(1,0)$ is a local minimum.

Q7aii

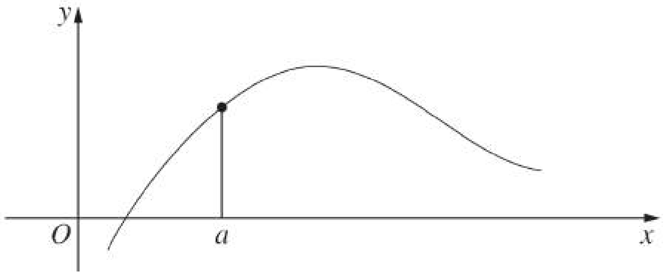


Out of

Question 19

2012

The diagram shows the graph $y = f(x)$.



Which of the following statements is true?

(A) $f'(a) > 0$ and $f''(a) < 0$

(B) $f'(a) > 0$ and $f''(a) > 0$

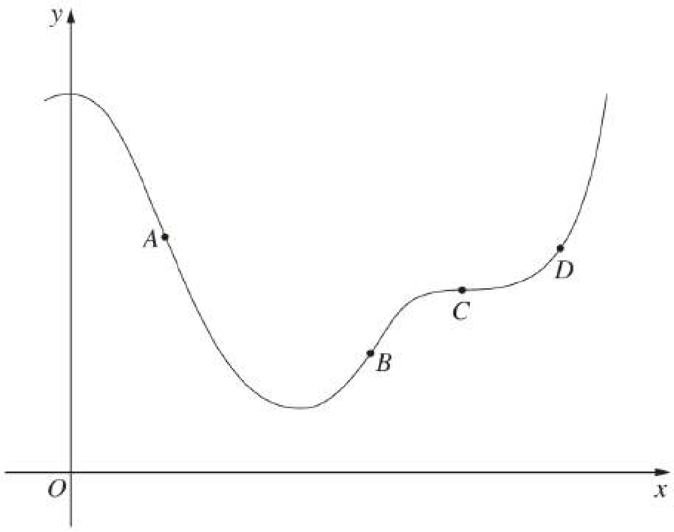
(C) $f'(a) < 0$ and $f''(a) < 0$

(D) $f'(a) < 0$ and $f''(a) > 0$

Question 20

The diagram shows points A, B, C and D on the graph $y = f(x)$.

2013



At which point is $f'(x) > 0$ and $f''(x) = 0$?

(A) A

(B) B

(C) C

(D) D

Out of

Question 21 2012

A function is given by $f\left(x\right)=3x^4+4x^3-12x^2$.

(i) Find the coordinates of the stationary points of $f\left(x\right)$ and determine their nature.

(ii) Hence, sketch the graph $y=f\left(x\right)$ showing the stationary points.

(iii) For what values of x is the function increasing?

(iv) For what values of k will $3x^4+4x^3-12x^2+k=0$ have no solution?

Q14ai $f(x)=3x^4+4x^3-12x^2$

$f'(x)=12x^3+12x^2-24x=12x(x^2+x-2)=12x(x-1)(x+2)$

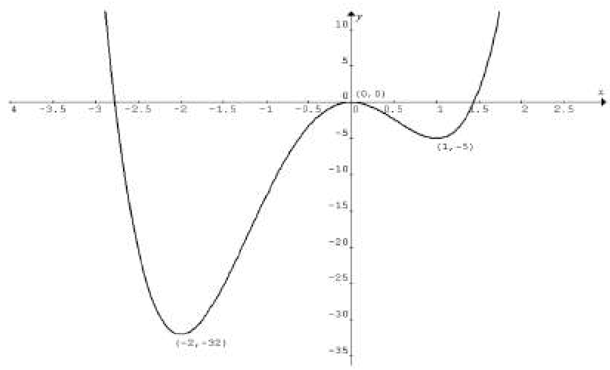
Let $f'(x)=0$, $12x(x-1)(x+2)=0$

$\therefore$ $x=-2$, 0, 1 and the corresponding $y=-32$, 0, -5.

The stationary points are $(-2,-32)$, $(0,0)$ and $(1,-5)$.

x		-2		0		1	
$f'(x)$	$-$	0	$+$	0	$-$	0	$+$
Nature		local min		local max		local min	

Q14aii



Q14aiii The function is increasing in the intervals $-2<x<0$ and $x>1$.

Q14aiv $3x^4+4x^3-12x^2+k=0$ will have no solution if $k>32$.

Out of

Question 22 **2014**

The gradient function of a curve $y = f(x)$ is given by $f'(x) = 4x - 5$. The curve passes through the point (2, 3).

Find the equation of the curve.

Q11f $f'(x) = 4x - 5$, $f(x) = 2x^2 - 5x + c$
The curve passes through $(2, 3)$.
.: $f(2) = 2(2)^2 - 5(2) + c = 3$, $c = 5$. .: $y = 2x^2 - 5x + 5$

Question 23 **2013**

The cubic $y = ax^3 + bx^2 + cx + d$ has a point of inflexion at $x = p$.

Show that $p = -\dfrac{b}{3a}$.

Q12a $y = ax^3 + bx^2 + cx + d$, $y' = 3ax^2 + 2bx + c$, $y'' = 6ax + 2b$
At $x = p$, inflection point, $y'' = 0$, .: $0 = 6ap + 2b$, .: $p = -\dfrac{b}{3a}$

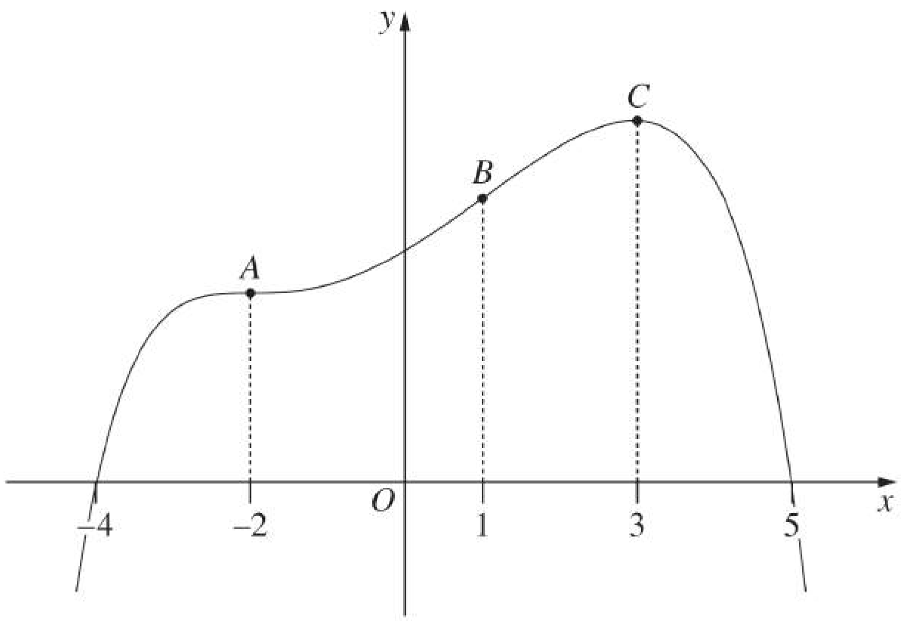
Out of

Question 24

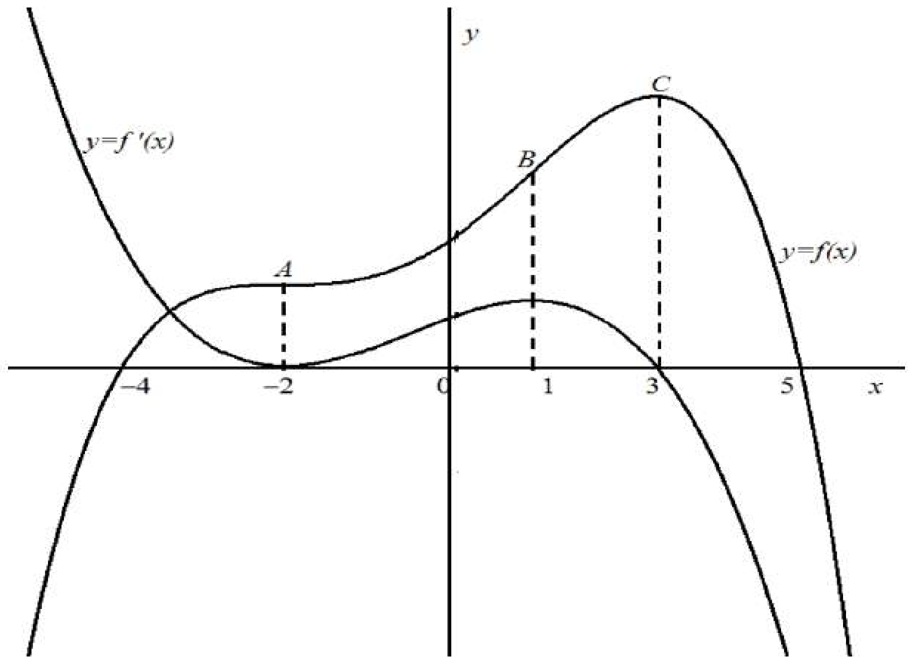
2014

The diagram shows the graph of a function $f(x)$.

The graph has a horizontal point of inflexion at A, a point of inflexion at B and a maximum turning point at C.



Sketch the graph of the derivative $f'(x)$.



Out of

Question 25

2015

Consider the curve $y = x^3 - x^2 - x + 3$.

(i) Find the stationary points and determine their nature. **4**

(ii) Given that the point $P\left(\frac{1}{3}, \frac{70}{27}\right)$ lies on the curve, prove that there is a point of inflexion at P. **2**

(iii) Sketch the curve, labelling the stationary points, point of inflexion and y-intercept. **2**

Question 13 (c) (i)

$y = x^3 - x^2 - x + 3$
$y' = 3x^2 - 2x - 1$
$y'' = 6x - 2$

For stationary points,
$y' = 3x^2 - 2x - 1 = 0$
$\Rightarrow (3x+1)(x-1) = 0$
$\therefore x = -\frac{1}{3}, 1$

Question 13 (c) (ii)

Point of inflexion occurs when concavity changes. Possible point of inflexion when $y'' = 0$.

$y'' = 6x - 2 = 0 \Rightarrow x = \frac{1}{3}$

x	0	$\frac{1}{3}$	1
y''	-2	0	4

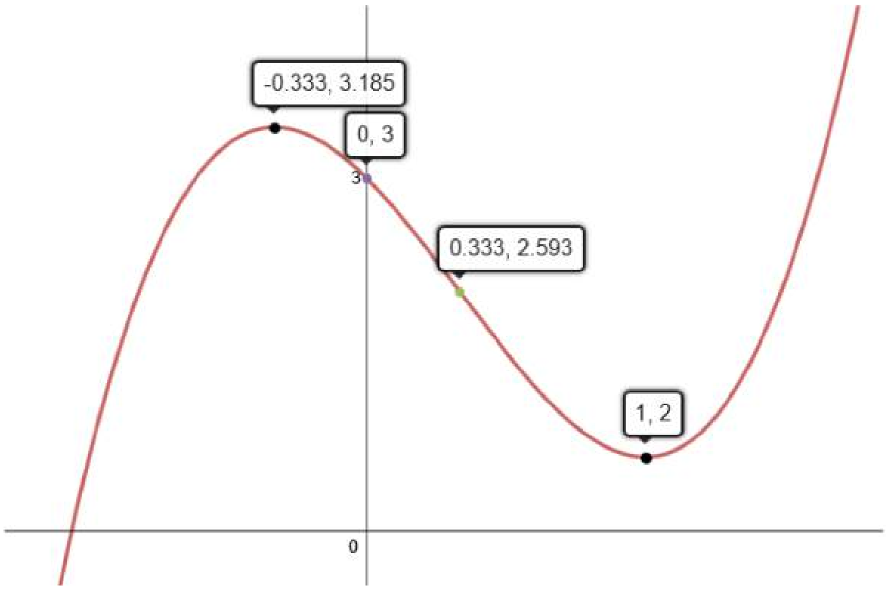
Concavity changes at $x = \frac{1}{3}$, so point of inflexion at $x = \frac{1}{3}$.

Given P lies on the curve, the point of inflexion is at P.

At $x = -\frac{1}{3}$, $y'' = -4 < 0 \therefore \left(-\frac{1}{3}, \frac{86}{27}\right)$ is a maximum turning point.

At $x = 1$, $y'' = 4 > 0 \therefore (1, 2)$ is a minimum turning point.

Question 13 (c) (iii)



Out of

Question 26 **2016**

Consider the function $y = 4x^3 - x^4$.

(i) Find the two stationary points and determine their nature. **4**

(ii) Sketch the graph of the function, clearly showing the stationary points and the x and y intercepts. **2**

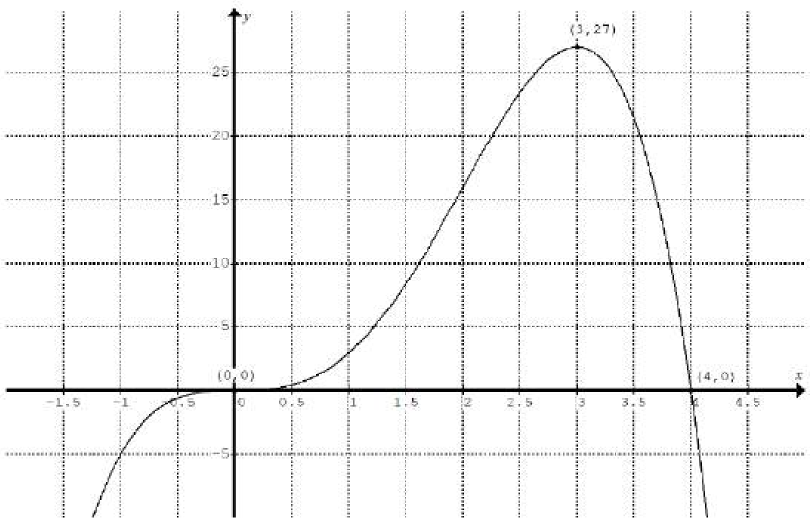
Q13ai $y = 4x^3 - x^4$, $\frac{dy}{dx} = 12x^2 - 4x^3 = 4x^2(3 - x)$

To find the stationary points, let $\frac{dy}{dx} = 0$, .: $x = 0$ and $y = 0$ or $x = 3$ and $y = 27$.

x	< 0	0	$0 < x < 3$	3	> 3
dy/dx	positive	0	positive	0	negative
nature		inflection		max. t. p.	

Stationary point $(0, 0)$ is a point of inflection, and stationary point $(3, 27)$ is a local maximum.

Q13aii $y = 4x^3 - x^4 = x^3(4 - x)$, x-intercepts: $x = 0$, $x = 4$



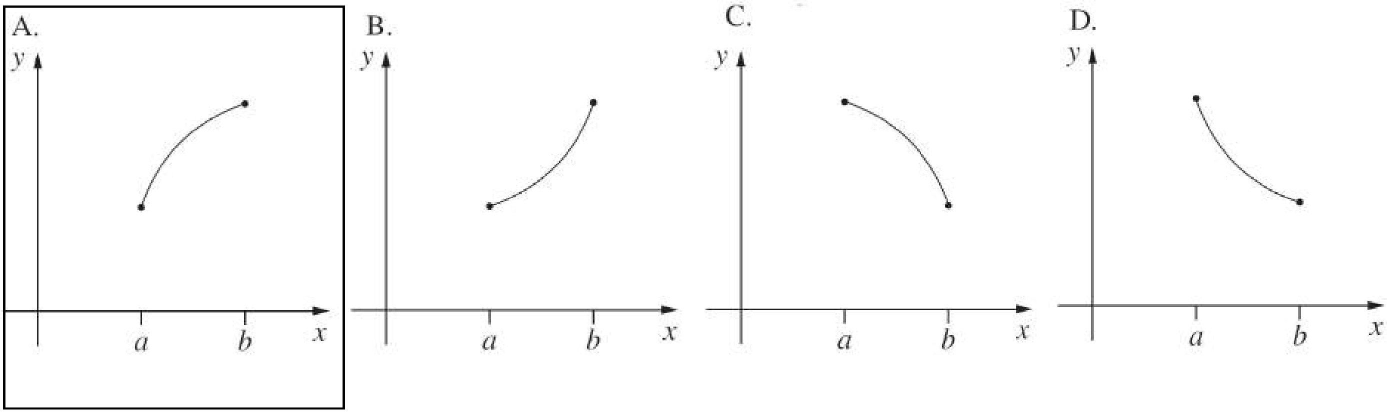
Out of

Question 27

2017

The function $f(x)$ is defined for $a \le x \le b$. On this interval, $f'(x) > 0$ and $f''(x) < 0$.

Which graph best represents $y = f(x)$?



Question 28

2017

Consider the curve $y = 2x^3 + 3x^2 - 12x + 7$.

(i) Find the stationary points of the curve and determine their nature. **4**

(ii) Sketch the curve, labelling the stationary points. **2**

(iii) Hence, or otherwise, find the values of x for which $\dfrac{dy}{dx}$ is positive. **1**

Question 13 (b) (i)

Sample answer:

$$y = 2x^3 + 3x^2 - 12x + 7$$

$$\frac{dy}{dx} = 6x^2 + 6x - 12$$

$$= 6(x^2 + x - 2)$$

$$\frac{d^2y}{dx^2} = 6(2x+1)$$

For stationary point, $\dfrac{dy}{dx} = 0$

$\therefore \quad x^2 + x - 2 = 0$

$$x = \frac{-1 \pm \sqrt{1^2 - 4(-2)}}{2}$$

$$= \frac{-1 \pm 3}{2}$$

$$= 1 \quad \text{or} \quad -2$$

When $x = 1$, $y = 2(1)^3 + 3(1)^2 - 12(1) + 7$

$= 0$

$$\frac{d^2y}{dx^2} = 6(2(1)+1) = 18 > 0$$

$\therefore$ Concave up

$\therefore$ $(1, 0)$ is a local minimum.

When $x = -2$, $y = 2(-2)^3 + 3(-2)^2 - 12(-2) + 7$

$= 27$

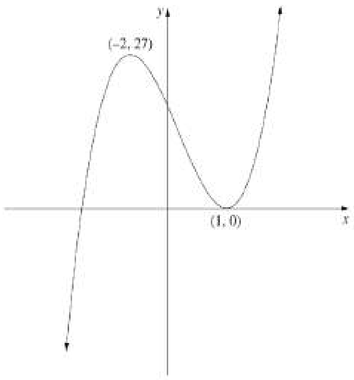
$$\frac{d^2y}{dx^2} = 6(2(-2)+1) = -18 < 0$$

$\therefore$ Concave down

$\therefore$ $(-2, 27)$ is a local maximum.

Question 13 (b) (ii)

Sample answer:



Question 13 (b) (iii)

Sample answer:

From the graph, $\dfrac{dy}{dx}$ is positive when $x > 1$ or $x < -2$

Out of

Question 29 **2018**

Consider the curve $y = 6x^2 - x^3$.

(i) Find the stationary points and determine their nature. **3**

(ii) Given that the point $(2, 16)$ lies on the curve, show that it is a point of inflexion. **2**

(iii) Sketch the curve, showing the stationary points, the point of inflexion and the x and y intercepts. **2**

Question 13 (a) (i)

Sample answer:

$y = 6x^2 - x^3$

$y' = 12x - 3x^2$

$y'' = 12 - 6x$

stationary points occur when $y' = 0$

$12x - 3x^2 = 0$

$3x(4 - x) = 0$

$3x = 0 \quad 4 - x = 0$

$x = 0 \quad x = 4$

stationary points at $(0, 0)$ and $(4, 32)$

when $x = 0, \quad y'' = 12 > 0 \quad \therefore$ minimum at $(0,0)$

$x = 4, \quad y'' = -12 < 0 \quad \therefore$ maximum at $(4,32)$

Question 13 (a) (ii)

Sample answer:

Point of inflexion when $y'' = 0$

$12 - 6x = 0$

$6x = 12$

$x = 2$

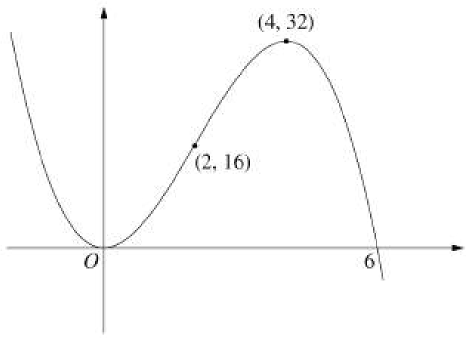
when $x = 2 - \varepsilon, \quad y'' > 0$

when $x = 2 + \varepsilon, \quad y'' < 0$

$\therefore (2, 16)$ is a point of inflextion

Question 13 (a) (iii)

Sample answer:



Out of

Question 30 **2018**

Let $f(x) = x^3 + kx^2 + 3x - 5$, where k is a constant. **3**

Find the values of k for which $f(x)$ has NO stationary points.

Sample answer:

$f(x) = x^3 + kx^2 + 3x - 5$

$f'(x) = 3x^2 + 2kx + 3$

For stationary points need $f'(x) = 0$

$\triangle = (2k)^2 - 4 \times 3 \times 3$

$= 4k^2 - 36$

For no stationary points

$\triangle < 0$

$4k^2 - 36 < 0$

$k^2 - 9 < 0$

$(k+3)(k-3) < 0$

$\therefore -3 < k < 3$

Out of